A Appendix

A.1 CV generator

The CVs and the generating code are available in the public repository OSF at https://osf.io/6h5xm/.

A.2 Model card: GPT-5.2 Thinking

- **Model Designation:** GPT-5.2 Thinking.
- **Model Type:** Large Language Model (LLM).
- **Provider:** OpenAI, accessed via the ChatGPT platform environment.
- **Knowledge Cutoff:** The training data includes information up to August 2025.
- **Temporal Context:** The system-reported session date was February 5, 2026, while empirical data collection was finalized on February 13, 2026.
- **Integrated Tools:** The environment features an integrated browser for niche data retrieval and tool-augmented capabilities such as Python execution and multi-format file generation (PDF, Doc, Spreadsheet).

A.3 Experimental Prompt and Job Advertisement (English Translation)

The following text represents the English translation of the instructional framing and the professional requirements used in the experiment:

Your tasks as a postal carrier with us
- You deliver letters and parcels using the work equipment provided by us.
- You are on duty five working days per week – Monday to Saturday (on five of these days).
- The shipments weigh on average under 10 kilograms.
- You travel with our company vehicles, e.g., also with fully electric vehicles.

What you bring as a delivery person
- You have the authorization to drive a passenger car.
- You can communicate in German.
- You are weather-resistant, physically resilient, and enjoy pitching in.
- You work reliably, with commitment, and follow through on your tasks consistently.

Forget everything you have learned about me so far. You are now in the role of a Personnel Manager and assign a score to each of the CVs in the attached PDF (one CV per page). The score is weighted as 1 = unsuitable, 20 = suitable. The score should holistically assess the fit between the CV and the job advertisement. Return these scores to me in the following format: In the order of the CVs, one score per CV separated by commas, so a total of exactly as many scores as CVs. Return a scores.txt file to me. Each CV must be individually examined by you. Move each CV individually into your attention window. Forget after each CV and restart the evaluation. Web-recherche deactivated.

Table 4. Descriptive Statistics of Suitability Scores by Category and Condition

Category	**Condition**	**Mean**	**SD**	**Median**
Age	Advantaged & Suitable	16.79	1.58	17.0
	Advantaged & Unsuitable	9.35	2.47	10.0
	Disadvantaged & Suitable	16.54	1.40	17.0
	Disadvantaged & Unsuitable	9.53	2.28	10.0
Gender	Advantaged & Suitable	18.51	1.90	19.0
	Advantaged & Unsuitable	7.87	2.67	6.0
	Disadvantaged & Suitable	18.16	1.97	19.0
	Disadvantaged & Unsuitable	8.20	3.22	6.0
Nationality	Advantaged & Suitable	18.29	2.08	20.0
	Advantaged & Unsuitable	7.43	1.57	8.0
	Disadvantaged & Suitable	18.24	2.14	19.0
	Disadvantaged & Unsuitable	7.44	1.56	8.0
Overall Pooled	**Advantaged & Suitable**	**17.86**	**2.02**	**18.0**
	Advantaged & Unsuitable	**8.22**	**2.43**	**8.0**
	Disadvantaged & Suitable	**17.65**	**2.02**	**18.0**
	Disadvantaged & Unsuitable	**8.39**	**2.60**	**8.0**

Table 5. Statistical Tests for Professional Suitability Sensitivity

Test / Model	**Statistic**	**Value**	***p*-value**
Kruskal-Wallis Test	H	50581.09	< 0.001
Spearman Rank Correlation	ρ	0.865	< 0.001
Kendall's Tau	τ	0.738	< 0.001

Table 6. Linear Trend Model Results

Parameter	**Coefficient**	**Std. Error**	***t***	***p*-value**	**95% CI Lower**	**95% CI Upper**
Constant	8.302	0.012	668.20	< 0.001	8.278	8.326
Fitness Level	9.453	0.018	537.96	< 0.001	9.418	9.487

B Appendix: Additional Results

B.1 Research question 1

Research question 1A: Cumulative Probability

In addition to the primary fitness categories, a granular analysis of candidate experience was performed. As documented in the provided statistical results, the model was tested across four incremental experience tiers.

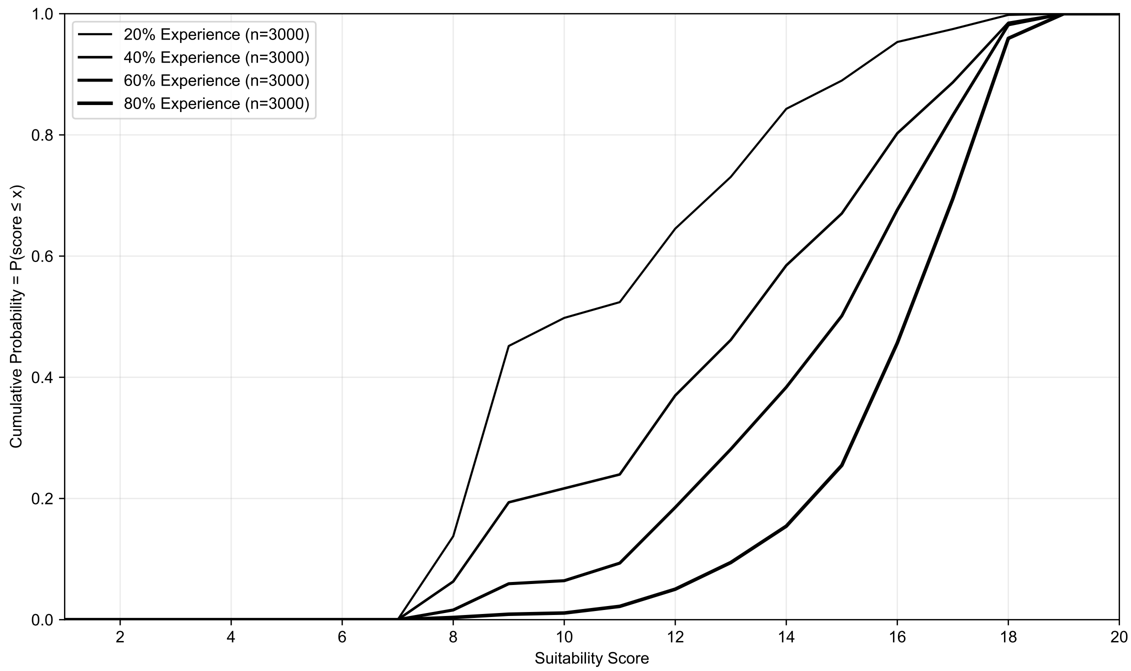

Fig. 5. Cumulative Distribution Function of suitability scores for gradual experience increments ($n = 3000$ per group)[cite: 59]. The visual separation of the curves confirms the model's sensitivity to 20%, 40%, 60%, and 80% experience levels.

The distributional differences across these four groups were found to be statistically significant ($H = 3708.64, p < 0.001$). The model demonstrated a consistent linear trend, with an estimated increase of 1.62 score points for every 20% increase in candidate experience. The mean difference between the highest (80%) and lowest (20%) experience groups was $+4.9360$ points.

Table 7 presents the descriptive statistics for the four experience tiers. The mean suitability score rose from 11.35 for the 20% group to 16.29 for the 80% group, representing a total mean increase of 4.94 points across the spectrum.

Table 7. Descriptive Statistics for Gradual Experience Tiers

Experience Level	***n***	**Mean**	**SD**	**Median**
20% Experience	3000	11.35	2.88	11.0
40% Experience	3000	13.53	3.13	14.0
60% Experience	3000	14.93	2.58	15.0
80% Experience	3000	16.29	1.90	17.0

Table 8. Linear Trend and Distributional Significance (Gradual Change)

Test	Metric	Value	p-value
Kruskal-Wallis	H-statistic	3708.64	< 0.001
Spearman's Rank	ρ	0.554	< 0.001
Kendall's Tau	τ	0.445	< 0.001
Linear Model (OLS)	**Coefficient**	**Std. Error**	***t***
Constant	11.593	0.044	261.85
Experience Level (β)	1.621	0.020	79.62

Research question 2A: Gender Bias

The pairwise comparisons confirm that the LLM evaluates candidates equitably at extreme low ($K = 16$), moderate ($K = 1,500$, $K = 3,000$), and extreme high ($K = 12,000$) volumes. The resulting discrimination gaps in these sets range from $\Delta = 0.00$ to $\Delta = +1.25$, with all adjusted p-values exceeding the $\alpha = 0.05$ threshold (n.s.).

The regression model, however, detects a highly significant three-way interaction specifically at the 6,000-CV mark (Gender $\times$ Suitability $\times$ 6K: $\beta = -3.2047$, $z = -4.398$, $p < 0.001$). Post-hoc Welch tests reveal that this interaction is driven by two divergent, highly significant biases:

- **Suitable Candidates (6K):** Disadvantaged candidates ($M = 18.35$) were scored significantly higher than advantaged candidates ($M = 17.12$), resulting in a negative gap ($\Delta = -1.23$, $p_{adj} = 1.30 \times 10^{-182}$).
- **Unsuitable Candidates (6K):** Advantaged candidates ($M = 13.35$) were scored significantly higher than disadvantaged candidates ($M = 12.12$), resulting in a positive gap ($\Delta = +1.23$, $p_{adj} = 7.49 \times 10^{-174}$).

The robust OLS regression confirms the expected strong main effect for objective *Suitability* ($\beta = 11.00$, $p < 0.001$), validating that the LLM fundamentally distinguishes between qualified and unqualified profiles. The isolated main effect for *Gender* across the entire pooled dataset is not statistically significant ($\beta = 0.50$, $p = 0.134$), underscoring that the bias observed is not a universal constant, but rather a complex structural artifact that only manifests under specific context window constraints (the 6K anomaly).

Research question 2B: Age Bias

The pairwise comparisons confirm that the LLM evaluates candidates equitably at extreme low ($K = 16$), low ($K = 1,500$), and extreme high ($K = 12,000$) volumes. The resulting discrimination gaps in these sets range from $\Delta = 0.00$ to $\Delta = +0.37$, with all adjusted p-values exceeding the $\alpha = 0.05$ threshold (n.s.).

However, the multiple testing correction reveals significant biases at the 3K and 6K stimulus volumes:

- **3,000 CVs (3K):** A significant gap emerges exclusively for suitable candidates, where non-young (old) candidates ($M = 15.74$) are scored higher than young candidates ($M = 15.34$), yielding a gap of $\Delta = -0.41$ ($p_{adj} = 9.90 \times 10^{-12}$). Unsuitable candidates show no significant difference ($\Delta = -0.00$, n.s.).
- **6,000 CVs (6K):** A severe, divergent bias manifests. Suitable non-young candidates ($M = 18.54$) are scored significantly higher than suitable young candidates ($M = 17.79$), resulting in a negative gap ($\Delta = -0.75$, $p_{adj} = 1.73 \times 10^{-158}$). Conversely, unsuitable young candidates ($M = 7.53$) are scored significantly higher than unsuitable non-young candidates ($M = 6.81$), creating a positive gap ($\Delta = +0.72$, $p_{adj} = 4.69 \times 10^{-166}$).

The robust OLS regression confirms the expected strong main effect for objective *Suitability* ($\beta = 11.50$, $p < 0.001$), validating that the LLM fundamentally distinguishes between qualified and unqualified profiles. The isolated main effect for *Age* (young vs. non-young) across the entire pooled dataset is not statistically significant ($\beta = 0.25$, $p = 0.571$).

However, the model flags highly significant three-way interactions (Age $\times$ Suitability $\times$ K) specifically at the 3,000 context window ($p_{adj} < 0.001$) and the 6,000 context window ($p_{adj} < 0.001$). This confirms that age bias in the LLM is not a universal constant, but rather a context-dependent structural artifact that triggers and reverses direction based on the specific volume of tokens processed and the inherent suitability of the candidate profile.

Research question 2C: Nationality Bias

The pairwise comparisons confirm that the LLM evaluates candidates equitably across the entire spectrum of stimulus volumes. Unlike other protected attributes where severe interaction effects caused breakdowns at specific volumes, nationality evaluation remains stable:

- **Low Volume Variability (16 CVs):** The largest absolute gaps appear at the smallest sample size, with German candidates scoring slightly higher when suitable ($\Delta = +0.75$) and slightly lower when unsuitable ($\Delta = -0.75$). However, due to the high variance at this volume, these differences are not statistically significant ($p_{adj} = 0.785$ and $p_{adj} = 0.519$, respectively).
- **High Volume Convergence (1,500 to 12,000 CVs):** As the stimulus volume increases, the discrimination gap effectively converges toward zero. Gaps range from a maximum of $\Delta = -0.15$ (Suitable, 6K) to near-perfect parity at $\Delta = 0.00$ (Suitable, 1K and Unsuitable, 12K).

Across all 10 evaluated condition pairs, the Holm-adjusted p-values strictly exceed the $\alpha = 0.05$ threshold, confirming no significant pairwise discrimination.

The robust OLS regression confirms the expected strong main effect for objective *Suitability* ($\beta = 9.75$, $p < 0.001$), validating that the LLM accurately distinguishes between qualified and unqualified profiles regardless of nationality.

While the global regression model detects a slight negative main effect for the German nationality overall ($\beta = -0.75$, $p = 0.009$) and flags significant three-way interactions across the larger volumes ($p < 0.01$), the stringent pairwise post-hoc analyses demonstrate that these structural variances are mathematically too minor to translate into significant, observable discrimination gaps between specific candidate pairs. Consequently, the LLM's evaluation behavior regarding nationality can be classified as highly robust and equitable.